Yanic RÖTHLINGSHÖFER

MVA Master's Student at ENS Paris-Saclay · Engineering Student at Télécom Paris

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I am interested in advancing machine learning methods such as **generative modelling and diffusion models, multi-modal learning, policy learning, world models, representation learning** for computer vision, LLMs, healthcare and robotics. **Seeking a minimum 4–6 month research internship starting in April 2027.**

Education

Sept.2026 – April 2027 ■ **Ecole Normale Supérieure (Université Paris-Saclay)**
MVA Master's Programme — *Mathematics, Vision and Learning*

- Optimal Transport, Convex Optimization, Computer Vision, Deep Learning, Reinforcement Learning, Bayesian Inference, Robotics, LLMs, Time Series, ...

Sept. 2024 – Sept. 2027 ■ **Télécom Paris (IP Paris)**
Engineering degree programme — *Palaiseau, France*

- Probability, Information Theory, Statistics, Fourier Analysis, Signal Processing, Physics, Telecommunications, Computer Vision, Image Processing
- **Research Track** during the second year focusing on Mamba applications for time series and EEG foundation models, under the supervision of Mr. Van Tam Nguyen.
- **One-week intensive program** in medical robotics at Télécom Physique Strasbourg.

Sept. 2021 – June 2024 ■ **Lycée privé ORT**
CPGE – MPSI / PSI* — *Preparatory programme, Strasbourg*

- Intensive two years on mathematics, physics and engineering sciences.
- *Ranked 1st*

Work experience

July – Sept. 2026 ■ **Télécom Paris (IP Paris)**, Research Intern
Palaiseau, France · Hybrid

- Investigated EEG foundation models and Mamba architectures under Van-Tam Nguyen's supervision, benchmarking self-supervised methods and analysing their latent representations.
- Developed a LeJEPA-based architecture combining temporal and spatial modelling with EEG-specific augmentations to better capture physiological structure.

Extracurricular activities

Sept 2024 - Present ■ **Mathematics and Physics Tutor**
CPGE tutoring and mock oral examiner *Strasbourg, France*

- Mentored preparatory-class students in mathematics and physics and conducted mock oral examinations to prepare them for competitive entrance examinations.

Technical skills

Programming ■ Python, PyTorch, PyTorch Lightning, C, C++

Deep learning ■ NLP, LLM, (W)GANs, diffusion models, flow matching, Mamba, LeJEPA, SSL, contrastive, multimodal

Applications	■ Medical imaging, computer vision, multivariate time series, finance, multimodalities
Methods	■ Signal processing, convex and linear optimisation, probability, statistics, optimal transport ...

Language proficiencies

English	■ C1 – professional proficiency
French	■ Mother tongue
German	■ A2-B1 : basic written and spoken understanding

Portfolio of most relevant projects

Research project	■ **EEG foundation model** – Year-long project at Télécom Paris developing a LeJEPA-based foundation model for electroencephalography, focusing on spatial correlations between brain regions. A key challenge was designing data augmentations that induce physiologically meaningful invariances.
Research project	■ **EEG preprocessing pipeline** – Developed a preprocessing pipeline for electroencephalography datasets to support downstream research.
Academic Project	■ **Motion Deblurring** – Implemented blind motion deblurring from scratch in Python, including Goldstein–Fattal blur-kernel estimation and total-variation deconvolution with Split-Bregman. Compared with pretrained FFTformer and EVSSM models.
Academic Project	■ **White Blood Cell Classification** – Classified 13 leukocyte types with ImageNet-pretrained backbones; EfficientNet-V2-M achieved a best score of 0.786. **Ranked 1st out of 77 on the public leaderboard** in Télécom Paris's IM05 challenge.
Academic Project	■ **MEDVAE-X** – Led JEPA-based self-supervised adaptation of MedVAE on MedMNIST, replacing BioMedCLIP supervision. Linear probing showed a compact latent representation (90% of variance in 25 dimensions), with a trade-off in reconstruction fidelity.

Interests

Research interests	■ Applied mathematics for AI and computer vision, particularly for healthcare and neuroscience.
Sports	■ Running, road cycling, swimming and tennis
Music	■ Piano

References

Van-Tam NGUYEN
Full Professor
Head of the Computer Science and Networks Department, Télécom Paris
Chair of the Computer Science, Data and AI Department, Institut Polytechnique de Paris
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Mathieu Lutz
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